Subject Description Form

Subject Code	AMA570
Subject Title	Current Topics in Actuarial Science
Credit Value	3
Level	6
Pre-requisite/ Co-requisite/ Exclusion	n/a
Objectives	Actuarial science is a field of study that applies knowledges in Mathematics and Statistics to assess and manage risks, particularly in the insurance and finance industries. This course equips students with the knowledges and practical techniques in the actuarial workplace environment. It also introduces the latest actuarial practices and regulations such as Hong Kong insurance regulation, International Financial Reporting Standard 17 (IFRS 17), Hong Kong Risk Based Capital (HKRBC).
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) Apply mathematical and statistical knowledge to solve practical problems encountered in the actuarial workplace. (b) Demonstrate an understanding of the latest standards and regulations specific to the insurance industry in Hong Kong. (c) Comprehend and apply the concepts and standards of international financial reporting relevant to actuarial science. (d) Exhibit professional integrity and ethical behaviour in the various disciplines associated with actuarial science. (e) Cultivate a commitment to lifelong learning for continuous industry advancement in the field of actuarial science.

Subject Synopsis/ Indicative Syllabus	The course covers the following major topics, with some additional topics and case-studies: **Hong Kong Insurance Regulation** Discussion of the regulatory Guidelines like below with illustrations of their applications and impacts in insurance company. - GL19 - Guideline on Qualifying Deferred Annuity Policy - Code of Practice for Insurance Companies under the Ambit of the Voluntary Health Insurance Scheme **Hong Kong Risk Based Capital (HKRBC)** Illustration of technical concept for calculation of HKRBC insurance liabilities, capital requirement and capital resources under the concept of Pillar 1 and Pillar 3 on disclosure requirements. **International Financial Reporting Standard 17 (IFRS 17)** Illustration for the basic concept of IFRS 17 under different models like General measurement model (GMM), Premium Allocation Approach (PAA) and Variable Fee Approach (VFA). **Application of Actuarial Practice** Practical illustration of actuarial practice in the working space like Product Pricing, Valuation, Financial Reporting and Experience analysis.:
Teaching/Learning Methodology	The subject will be delivered mainly through lectures. The teaching and learning approach mainly focus on the latest actuarial practice. The approach aims at the development of concept on how to apply actuarial concept and framework into the daily practical problems in actuarial profession. Students are encouraged to adopt a deep study approach by employing high level cognitive strategies, such as critical and evaluative thinking, relating, integrating and applying theories to practice.
Assessment Methods in Alignment with Intended Learning Outcomes	(see table and notes below)

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
		a	b	c	d	e
In-class Exercise / Discussion	10%	✓	✓	✓		✓
Assignment	10%	✓	✓			✓
Mid-term Test	40%	✓	✓	✓		
Final Presentation	40%		✓	✓	✓	✓

- The **In-class Exercise / Discussion** requires students to participate in interactive exercises and discussions to reinforce key concepts, apply theory to real-world scenarios, and develop critical thinking and collaboration skills in a dynamic, practical learning environment in the lecture.
- The **Assignment** requires students to complete a practical task based on the professor's instructions. Its purpose is to simulate real-world professional tasks.

	• The **Midterm Test** contains a set of questions based to the knowledge covered in the course. It assesses students' understanding of concepts, practical techniques and problem-solving skills in actuarial science and related disciplines such as insurance and finance. • The **Final Presentation** requires students to present some case study problems that illustrates their understanding of the concepts and application of the course knowledge.

Student Study Effort Required	Class contact:	
	▪ Lecture	39 Hrs.
	Other student study effort:	
	▪ Readings and exercises	20 Hrs.
	▪ Final Presentation Preparation	20 Hrs.
	▪ Self-study	30 Hrs.
	Total student study effort	109 Hrs.

Reading List and References

GL19 - Guideline on Qualifying Deferred Annuity Policy	The Insurance Authority
Code of Practice for Insurance Companies under the Ambit of the Voluntary Health Insurance Scheme	Health Bureau
Hong Kong RBC – Second Quantitative Impact Study (QIS 2)	Milliman, Inc
Hong Kong RBC – Third Quantitative Impact Study (QIS 3)	Milliman, Inc
IFRS 17 Insurance Contracts Standard	The International Accounting Standards Board